XAVI BOSCH

INTERACTIVE PRODUCT DEVELOPMENT · JUNIOR PROFILE

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PROFESSIONAL PROFILE

Second year student of the Interactive Product Design and Creation degree at La Salle URL and a self taught developer, with ten products built end to end and several deployed and in daily use. Main stack in Python, Swift and TypeScript on React, with Firebase as the backend and computer vision through OpenCV and MediaPipe. I write the data model and the interface at the same time, and I count nothing as verified until it runs on the real device.

SELECTED PROJECTS

Betsy `github.com/xavibosch/betsy-ios`

iOS app for social betting between friends, on virtual points with nothing convertible to money.

STACK: **Swift, SwiftUI, Cloud Firestore, Firebase Security Rules, Xcode**

- Designed the data model and wrote the Firestore security rules alongside the SwiftUI views, which exposed a race condition letting two users settle the same bet. Moved write authority to the server and left the interface able only to request.
- Defined the target users, mapped the customer journey and ran tests with real users before building the screens, then reshaped the interface from the results rather than from my own taste.

Gesture TV `github.com/xavibosch/gesture-tv`

Full television control by gesture and voice, for a person who cannot use a remote.

STACK: **Python, OpenCV, MediaPipe, Vosk, Wake on LAN**

- Developed a real time computer vision pipeline with OpenCV and MediaPipe that reads the hand by screen zone, plus offline speech recognition with Vosk for the remaining commands.
- Redesigned the architecture to drop full shell access and replace it with Wake on LAN and a narrow command set, cutting the attack surface. No camera frame is ever written to disk.

Reps `github.com/xavibosch/reps`

Gym dashboard driven by a hand in the air, with no screen to touch between sets.

STACK: **React, TypeScript, Vite, MediaPipe, Electron, Spotify Web API**

- Built the hand tracking and a two state machine, active and standby, in React. Packaged it as a desktop application with Electron and integrated the Spotify API for playback control.
- Diagnosed a gap of several centimetres between the hand cursor and the hit test. An ancestor carrying a CSS transform becomes the containing block for position fixed descendants, so the cursor was laid out in the scaled frame while the hit test read the real viewport. Rendering it outside that frame put both back in one coordinate system.

TECHNICAL STACK

Languages: Python, Swift, TypeScript, JavaScript, Java, HTML, CSS

Frameworks and libraries: SwiftUI, React, Electron, Vite, OpenCV, MediaPipe, Vosk, Core Haptics, PWA

Tools, databases and cloud: Git, GitHub, Firebase with Cloud Firestore, Realtime Database and Security Rules, Vercel, MQTT, Ollama, Xcode, Figma, Arduino, ESP32

EDUCATION AND CERTIFICATIONS

BSc in Interactive Product Design and Creation La Salle, Universitat Ramon Llull, Barcelona. 2025 to present.
First year complete, 60 ECTS in programming, object orientation, UML, electronics and interface design. Second year from September 2026.

English C1 advanced

Self taught development
Ten products of my own designed and built end to end, across iOS apps, web apps and systems that pair software with hardware.

KEY COMPETENCIES

Logical problem solving: I isolate components before integrating them. Debugging an assembled system with motors, sensors and radio all live cost hours, and testing each part alone found the fault in minutes.

Autonomous learning: I learned the whole stack above because a project needed it, with no course in between.

Teamwork and interface design: On a paired project we settled the contracts between layers before writing the code behind them, which removed the mutual blocking.

Product judgement: I delete a feature when it is ambiguous by construction rather than keep tuning it.